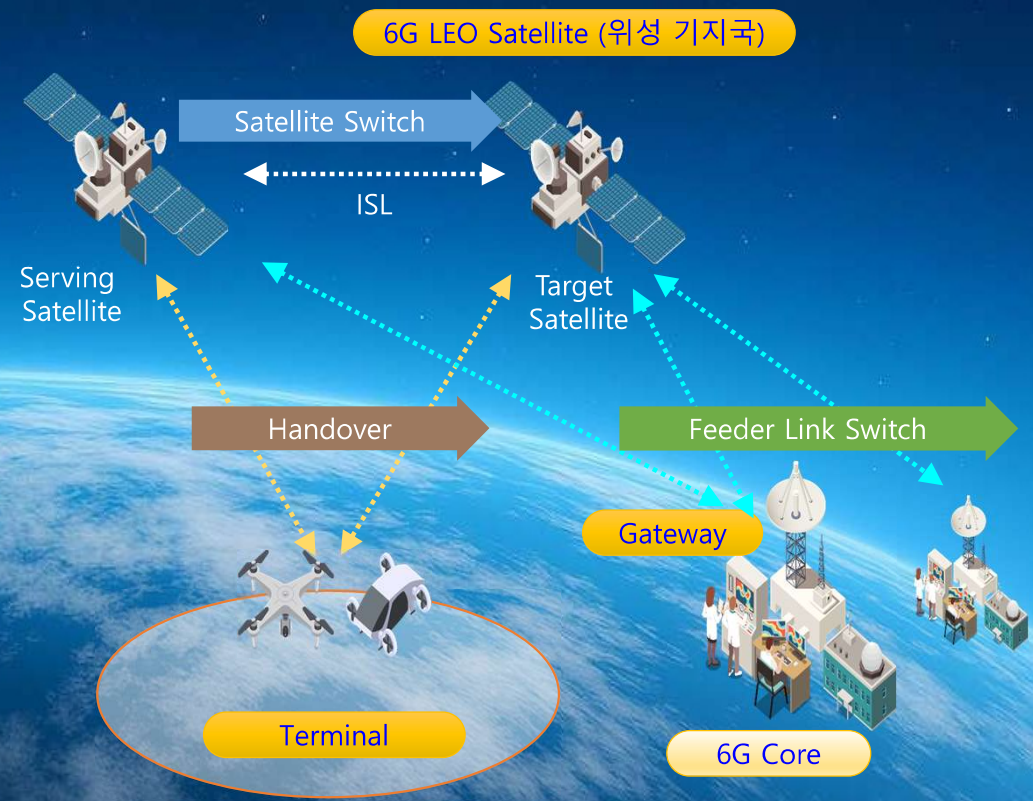


1-3 서비스 운용 개념

단말 등록/세션 설정을 통한 서비스 제공 시나리오

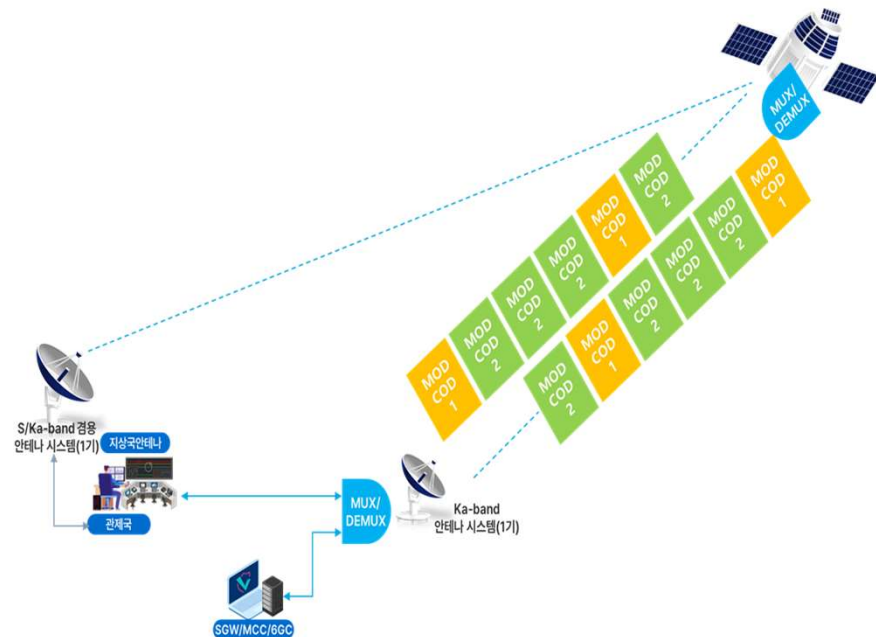


단말/위성 이동에 따른 이동성 지원 시나리오



3-1 링크 버짓 - TM/TC

- ▶ [SR500080] Ka-band In-band TM/TC 분석결과
- ▶ S-band TM/TC 분석결과



<S & Ka-band TM/TC 운용 개념(안)>

항목	단위	값
지상국 안테나 크기	m	5.4
HPA	W	100
전송방식 (CCSDS)	변조방식	- QPSK
	FEC	- RS & CC
최소 양각	degree	10

<S-band 지상국 제원(TBC)>

항목	단위	마진
링크마진	DL	dB 4.13
	UL	dB 12.83

<S-band TM/TC 링크분석 결과(TBC)>

항목	단위	값
지상국 안테나 크기	m	2.9
HPA	W	100
전송방식 (DVB-S2X SF)	변조방식	- QPSK
	FEC	- 1/2
최소 양각	degree	15*

* FL 안테나 운용각 고려한 SGW에서의 양각

<Ka-band 지상국 제원(TBC)>

항목	단위	마진
링크마진	DL	dB 5.2
	UL	dB 12.5

<Ka-band In-band TM/TC 링크분석 결과(TBC)>

3-2 링크 버짓 - 피더링크

▶ [SR201050/SR301010] 100MHz 대역폭 최대 400Mbps 전송속도 만족 확인

지상국 피더링크 제원	단위	값
대역	GHz	20
수신 안테나 크기	m	2.9
효율		0.56
안테나 이득	dB	2.5
피더 손실	dB	2.5
LNB 잡음지수	dB	1.6
G/T	dB/K	27.2

<지상국 피더링크 수신안테나 제원 및 G/T>

Ka-band feeder/downlink (Sat->Gat)		
Freq. band	20	GHz
Occupied BW	100	MHz
EIRP	40.0	dBW
Altitude	888	km
Elevation angle	30	deg
Distance	1532	km
L_free	182.2	dB
L_other	1	dB
GW G/T	27.2	dB/K
k	-228.6	dBW/K/Hz
C/N	32.59	dB
Roll-off factor	0.05	
Symbol rate	95.24	MHz
2^modulation order	64	
Code rate	0.83	
Es/N0 threshold	16.55	
spectral efficiency	4.9366	bits/symbol
Data rate	470.15	Mbps
Eb/N0	25.86	dB
Es/N0	32.80	dB
Link margin	16.25	dB
Data rate (w/o overhead)	423.14	Mbps

<피더 링크 링크버짓: 통신위성 → SGW>

3-3 링크 버짓 - 서비스링크

▶ [SR201060] 100MHz 대역폭 최대 300Mbps 전송속도 만족 확인

서비스링크 제원	단위	값
소자 방향성 이득 (0.58l, 80%)	dBi	4.3
총 방향성 이득	dBi	32.68
급전 손실	dB	2
복사 소자 출력	dBm	19
총 RF 출력 (dBW)	dBW	17.38
총 RF 출력 (linear)	W	54.65
EIRP	dBW	48.1

<위성 서비스링크 송신안테나 제원 및 EIRP>

Ka-band service/downlink		
freq. band	20	GHz
Occupied BW	100	MHz
EIRP(Max)	48.1	dBW
EIRP(Per beam)	39.1	dBW
EIRP(Mean)	31.1	dBW
altitude	888	km
elevation angle	90	deg
distance	888	km
L_free	177.4	dB
L_other	1	dB
Terminal G/T	13.5	dB/K
k	-228.6	dBW/K/Hz
C/N	14.76	dB
# of RBs	66	
# of REs per RB	12	
Numerology	3	
Num. of symbols per slot	14	
Symbol rate	112000	
CP length	0.57	us
symbol duration(w/o CP)	8.33	us
symbol duration	8.92	us
SCS	120	kHz
Symbol BW	95.04	MHz
Es/N0	14.98	dB
Eb/N0	9.41	dB
MCS Index number	14	
2^modulation order	64	
Target coderate*1024	616	
code rate	0.602	
Target BLER	0.01	
Es/N0 threshold	12.63	
spectral efficiency	3.6094	bits/symbol
data rate (w/ overhead)	316.96	Mbps
link margin	2.35	dB
Overhead	2.00	%
data rate (w/o overhead)	310.63	Mbps

<서비스 링크 링크버짓: 통신위성➡ 단말>

3. 시스템 요구사항 분석

3-6 Link Budget Analysis (UR40020, SR401030, SR402010)

➤ 본 Link Budget은 SR401030(Communication Frequency) 및 SR402010(User link Capability), UR40020(채널 전송속도) 요구사항을 만족하기 위한 설계 근거이며, Ka band에서 EIRP(per Beam) Level 및 Back-off를 포함한 DL/UL Throughput을 산출함

• Link Budget Analysis(User link)

- DL TP with 8 dB Back-off from 39 dBW per beam in payload is about 200 Mbps
- The linear EIRP for 64-QAM for 300 Mbps should be at least 38 dBW which is Pout of payload in DL
- UL TP target of 40 Mbps can be achieved with 20 MHz BW

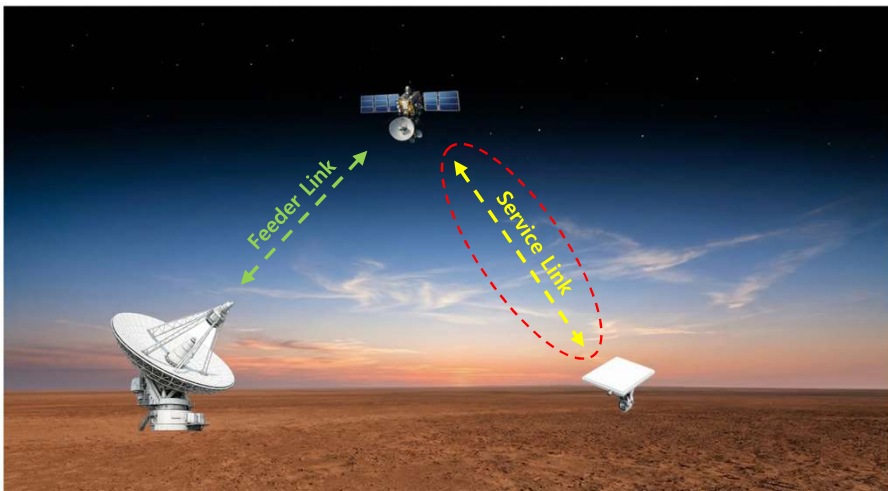
- ✓ Target DL TP: 300 Mbps
- ✓ Target UL TP: 40 Mbps

Downlink	Parameter	Unit	Value
	Center freq	GHz	20
	DL channel BW	MHz	100
	Control Channel Overhead	-	0.18
	Range at boresight (90 deg)	km	888
	Elevation Angle (90- Scan range)	deg(+/-)	39
	Slant range	km	1,299
	n of beams	n	8
	Peak EIRP (composited 8-beam and single-pol)	dBW	48.1
	Peak EIRP per beam	dBW	39.1
	Back-off for SNR 28dB (4% EVM for 64QAM)	dB	1.0
	Composited EIRP with Back-off	dBW	47.1
	Linear EIRP per beam with Back-off	dBW	38.1
	Path loss dB @ boresight	dB	177.4
	Path loss dB @ max slant	dB	180.7
	Atmospheric loss	dB	1.0
	Scintillation loss	dB	0.0
	Rain attenuation	dB	0.0
	G/T	dB/K	13.9
	System Noise Temperature	K	183.9
	Sky temperature	K	35.0
	Ground terminal NF	dB	1.8
	Coherent gain (unit-cell gain+ BF Gain)	dB	37.0
	BF Error	dB	0.5
	Terminal EIS (required SNR: 0 dB)	dBm/BW(Hz)	-129.2
	SNR (noise limited)	dB	18.8
	SNR (Non-linearity limited)	dB	28
	SNR (total)	dB	18.3
	SNR (noise limited) at max slant angle	dB	15.5
	SNR (NL limited) at max slant angle	dB	28
	SNR (total) at max slant angle	dB	15.3
	[max] Optimum spectral efficiency from MCS table for the calculated SNR	b/s/Hz	4.2
	[min] Optimum spectral efficiency from MCS table for the calculated SNR	b/s/Hz	3.6
	Max. DL Throughput	Mbps	306.4
	Min. DL Throughput	Mbps	262.5

Uplink	Parameter	Unit	Value
	Center freq	GHz	30
	UL channel BW	MHz	20
	Control Channel Overhead	-	0.18
	Range at boresight (90 deg)	km	888
	Elevation Angle (90- Scan range)	deg(+/-)	30
	Slant range	km	1,532
	n of beams	n	1
	Peak EIRP (composited 8-beam and single-pol)	dBW	49.0
	Peak EIRP per beam	dBW	49.0
	Back-off for SNR xx dB (x% EVM for 16QAM)	dB	8.0
	Composited EIRP with Back-off	dBW	41.0
	Linear EIRP per beam with Back-off	dBW	41.0
	Path loss dB @ boresight	dB	181.0
	Path loss dB @ max slant	dB	185.7
	Atmospheric loss	dB	1.0
	Scintillation loss	dB	0.0
	Rain attenuation	dB	0.0
	G/T	dB/K	5.0
	System Noise Temperature	K	199.6
	Sky temperature	K	30.0
	Ground terminal NF	dB	2
	Coherent gain (unit-cell gain+ BF Gain)	dB	28.5
	BF Error	dB	0.5
	Payload EIS (required SNR: 0 dB)	dBm/BW(Hz)	-127.5
	SNR (noise limited)	dB	16.5
	SNR (Non-linearity limited)	dB	28
	SNR (total)	dB	16.2
	SNR (noise limited) at max slant angle	dB	11.8
	SNR (NL limited) at max slant angle	dB	28
	SNR (total) at max slant angle	dB	11.7
	[max] Optimum spectral efficiency from MCS table for the calculated SNR	b/s/Hz	3.6
	[min] Optimum spectral efficiency from MCS table for the calculated SNR	b/s/Hz	2.6
	Max. UL Throughput	Mbps	52.5
	Min. UL Throughput	Mbps	37.3

3. 시스템 요구사항 분석

Link Budget 시뮬레이션 결과 - Uplink



Service link 주요 재원

Tx EIRP : 43 dBW(6dB Back-off)
Rx G/T : 5 dB/K
Distance : 888Km (Boresight 기준)

Frequency : 30 GHz
Bandwidth : 50 MHz
Bit Rate : 40 Mbps
Modulation : 16QAM

Uplink 시뮬레이션 결과

Bit Rate : 40Mbps
Modulation : 16 QAM
Required Eb/No : 13.5 dB
Link Margin : 2 dB 확보
(Boresight 기준)

Satellite & Receiver

Ground Station	Link	Satellite
Name S1		
Type Satellite		
PS1	Latitude (deg)	37.5
PS2	Longitude (deg)	127.5
PS3	Altitude (km)	888.02
Transmitter		
None		
▼ Receiver		
PR1	Interference loss (dB)	0
PR2	Rx G/T (dB/K)	5

Link

Ground Station	Link	Satellite
Name L1		
Type Link		
From G1		
To S1		
PL1	Frequency (GHz)	30
PL2	Bandwidth (MHz)	50
PL3	Bit rate (Mbps)	80
PL4	Required Eb/No (dB)	13.6
PL5	Polarization mismatch (deg)	0
PL6	Implementation loss (dB)	1

ESA & Transmitter

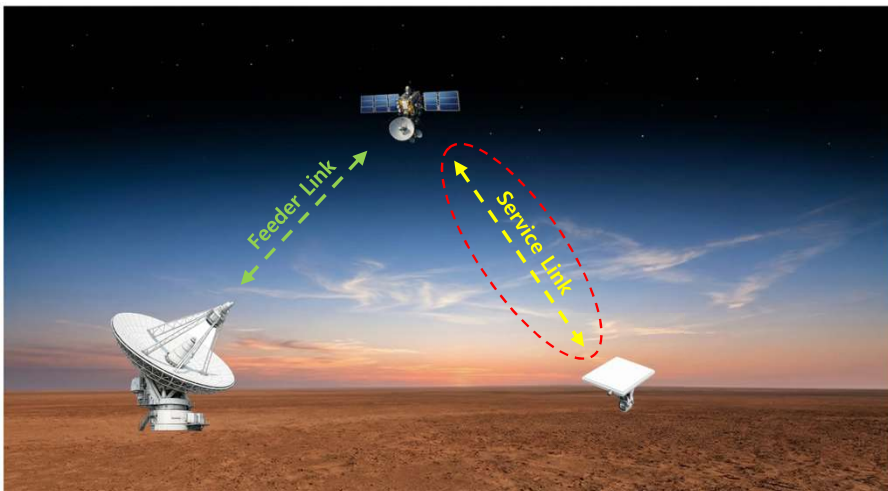
Ground Station	Link	Satellite
Name G1		
Type Ground Station		
PG1	Latitude (deg)	37.5
PG2	Longitude (deg)	127.5
PG3	Altitude (m)	20
▼ Transmitter		
PT1	Tx feeder loss (dB)	0
PT2	Other Tx losses (dB)	0
PT3	Tx HPA power (dBW)	40
PT4	Tx HPA OBO (dB)	6
PT5	Tx antenna gain (dBi)	9

Uplink Link Budget 시뮬레이션 결과

Tag	Name	Uplink
N1	Distance (km)	888.0000
N2	Elevation (deg)	90.0000
N3	Tx EIRP (dBW)	43.0000
N4	Polarization loss (dB)	0
N5	FSPL (dB)	180.9585
N6	Received isotropic power (dBW)	-137.9585
N7	C/No (dB-Hz)	95.6407
N8	C/N (dB)	18.6510
N9	Received Eb/No (dB)	16.6098
N10	Margin (dB)	2.0098

3. 시스템 요구사항 분석

Link Budget 시뮬레이션 결과 - Downlink



Service link 주요 자원

Tx EIRP : 38 dBW(1dB Back -Off)
Rx G/T : 13.9 dB/K
Distance : 888Km (Boresight 기준)

Frequency : 20 GHz
Bandwidth : 100 MHz
Bit Rate : 300 Mbps
Modulation : 64QAM

Downlink 시뮬레이션 결과

Bit Rate : 300Mbps
Modulation : 64 QAM
Required Eb/No : 18 dB
Link Margin : 0.39 dB 확보
(Boresight 기준)

ESA & Receiver

Ground Station	Link	Satellite
Name G2		
Type Ground Station		
PG1	Latitude (deg)	37.5
PG2	Longitude (deg)	127.5
PG3	Altitude (m)	20
Transmitter		
None		
▼ Receiver		
PR1	Interference loss (dB)	0
PR2	Rx G/T (dB/K)	13.9
PR3	Rx feeder loss (dB)	0
PR4	Other Rx losses (dB)	0

Link

Ground Station	Link	Satellite
Name L2		
Type Link		
From S2		
To G2		
PL1	Frequency (GHz)	20
PL2	Bandwidth (MHz)	100
PL3	Bit rate (Mbps)	300
PL4	Required Eb/No (dB)	18
PL5	Polarization mismatch (deg)	0
PL6	Implementation loss (dB)	1

Satellite & Transmitter

Ground Station	Link	Satellite
Name S2		
Type Satellite		
PS1	Latitude (deg)	37.5
PS2	Longitude (deg)	127.5
PS3	Altitude (km)	888.02
▼ Transmitter		
PT1	Tx feeder loss (dB)	0
PT2	Other Tx losses (dB)	0
PT3	Tx HPA power (dBW)	1
PT4	Tx HPA OBO (dB)	1
PT5	Tx antenna gain (dBi)	39.1

Downlink Link Budget 시뮬레이션 결과

Tag	Name	Downlink
N1	Distance (km)	888.0000
N2	Elevation (deg)	90.0000
N3	Tx EIRP (dBW)	39.1000
N4	Polarization loss (dB)	0
N5	FSPL (dB)	177.4366
N6	Received isotropic power (dBW)	-138.3366
N7	C/No (dB-Hz)	104.1625
N8	C/N (dB)	24.1625
N9	Received Eb/No (dB)	19.3913
N10	Margin (dB)	0.3913

3. 시스템 요구사항 분석

변조방식 별 Required Eb/N0

변조 방식 (Modulation)	Bits / Symbol	Required Eb/N0 (BER 10 ⁻³ 목표)	Required Eb/N0 (BER 10 ⁻⁶ 목표)
BPSK	1	6.8 dB	10.5 dB
QPSK	2	6.8 dB	10.5 dB
8-PSK	3	10.0 dB	14.0 dB
16-QAM	4	10.5 dB	14.4 dB
64-QAM	6	14.8 dB	18.8 dB
256-QAM	8	19.0 dB	23.0 dB